Adify Skills
n8n Automation Mastery

📘 CLASS NOTES

🤖 Module 5 · AI Automation & Agents

DAY 15 of 25

AI Agents: Tools, Memory & Reasoning

“Chatbot nahi — ek digital employee banao.”

⚡ AAJ HUM YEH BANAYENGE

Telegram AI Assistant (Agent + Tools + Memory)

Ek personal AI assistant Telegram par: "kal 4 baje Rahul ke saath meeting rakho", "₹450 lunch expense likho", "18% GST of 12500?" — agent khud decide karta hai Calendar, Sheet ya Calculator kaunsa tool use kare, aur pichhli baat yaad rakhta hai.

AI Agent | Chat Trigger | Window Memory | Tools

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

AI Agent = LLM + tools + memory. Yahi "digital employee" hai jo clients sabse zyada maangte hain.

🎯 Aaj ke baad aap yeh kar paoge

01 AI Agent node: how agents think & act

🌍 Didi pehle sochti hai, phir karti hai.
⚙️ AI Agent: Think → Tool call → Observe → Answer.

02 Tools: HTTP, Calculator, Code, sub-workflow as tool

🌍 Didi ke paas phone, calculator, diary.
⚙️ Tools: HTTP Request, Calculator, Code, Call n8n Workflow.

03 Memory: window buffer, Postgres/Redis chat memory

🌍 Didi ko yaad hai tumhe chai pasand hai.
⚙️ Window Buffer Memory (last 10 msgs), Postgres memory (permanent).

04 System prompts for reliable agents

🌍 Didi ko job description diya.
⚙️ System prompt: role, rules, tools kab use karein, kya nahi karna.

05 Chat Trigger & embedding chat on websites

🌍 Dukaan ke counter par help desk.
⚙️ Chat Trigger → embed widget website par.

06 Agent guardrails & human-in-the-loop

🌍 Bade kharche se pehle papa se poochna.
⚙️ Payment/delete se pehle Slack/Telegram approval.

⏱ Class ka plan (2 ghante)

10m	35m	45m	20m	10m

0:00-0:10 Recap

🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.

0:10-0:45 Concept: Agents

ReAct loop visual

🪄 Chatbot sirf baat karta hai. Agent ke paas haath bhi hain — woh calendar khol sakta hai, calculator chala sakta hai.

0:45-1:30 Live Build

Personal assistant agent: Calendar + Gmail + Sheets tools

🪄 Aaj ki kahani yaad hai — 'Personal assistant didi'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.

1:30-1:50 Challenge

Add a custom tool via sub-workflow

🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.

1:50-2:00 Q&A + Homework

🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Agent loop

Socho → tool chuno → result dekho → dobara socho → jawab.

🪄 **Analogy:** Detective jo clues dhoondhta hai

CONCEPT 02

Tools

Agent ke haath-pair: API, calculator, sub-workflow.

🪄 **Analogy:** Mechanic ka toolbox

CONCEPT 03

Memory

Pichhli baatein yaad rakhna (session ID se).

🪄 **Analogy:** Dukaan wala jo regular customer ko pehchaanta hai

CONCEPT 04

System prompt

Agent ki job description + rules + guardrails.

🪄 **Analogy:** Company ki rulebook

🔑 Aaj ka main idea

AI Agent ek LLM hai jise tools diye gaye hain. Woh khud decide karta hai kaunsa tool, kis order mein chalana hai, aur memory se pichhli baat yaad rakhta hai.

- Telegram bot jo expense Sheet mein likhe
- Website chat jo order status bataye
- Sales agent jo CRM se lead info nikaale

📖 Ek kahani se samjho: Personal assistant didi

1 📱 **Tumne message kiya "kal meeting set karo"** = Chat/Telegram Trigger

2 🤔 **Didi ne socha kya karna hai** = AI Agent (reasoning)

3 📅 **Calendar khola** = Tool: Google Calendar

4 🧠 **Pichhli baat yaad rakhi** = Memory

5 ✅ **"Ho gaya, 4 baje set hai"** = Reply

🪄 Ek line mein: “Chatbot sirf baat karta hai. Agent ke paas haath bhi hain — woh calendar khol sakta hai, calculator chala sakta hai.”

🗺️ Aaj ka workflow: Telegram AI Assistant (Agent + Tools + Memory)

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

Busy founders, CA firms (AI personal assistant) — aam taur par **₹15,000 – ₹40,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1 ⚡ **Telegram Trigger**
User message

2 🤖 **AI Agent**
System prompt: personal assistant, Hinglish, confirm before booking

3 🧠 **OpenAI Chat Model**
Agent ka model (sub-node)

4 💾 **Simple Memory (Window Buffer)**
Session key = chat ID, last 10 messages

5 🧩 **Call n8n Workflow Tool**
Tools: book_meeting, log_expense, summarize_inbox

6 ✈️ **Telegram**
Agent ka jawab wapas

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- [ ] Connect: **Telegram bot token (BotFather se)**
- [ ] Connect: **OpenAI API key**
- [ ] Connect: **Google Sheets (Google OAuth)**
- [ ] Connect: **Google Calendar (Google OAuth)**
- [ ] "Expenses" tab headers: date, amount, category, note
- [ ] Workflow Active karo

✅ Test kaise karein

1. "Lunch 450 rupaye" bhejo → Sheet
2. "Kal 4 baje Rahul se call" → Calendar
3. "Aur uska reminder kab hai?" — memory test

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😫 Pehle (manual)

Calendar check, slot dhoondhna, invite bhejna — **10 min**

10 min

⚡ Ab (n8n)

Telegram par ek line — **1 min**

1 min

⏱ Din mein 10 aise kaam = ~1.5 ghante bache.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo — 📋 Select karke copy karo

```
You are Aria, a personal assistant.
- Always confirm date/time before booking.
- Use tools; never invent data.
- Reply in short Hinglish.
```

📦 Sample data — input se output tak

📥 Input — 📋 Select karke copy karo

```
{
  "text": "Kal 5 baje Rohan ke saath meeting book karo aur aaj ka ₹450 lunch expense log karo"
}
```

📤 Output — 📋 Select karke copy karo

```
{
  "reply": "✅ Kal 5:00 PM Rohan ke saath meeting book ho gayi. ₹450 lunch expense log kar diya.",
  "toolsUsed": [
    "book_meeting",
    "log_expense"
  ]
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Agent system prompt

📋 Select karke copy karo

```
Tum mera personal assistant ho. Aaj ki date: {{today}}, timezone Asia/Kolkata.
Rules:
- Meeting ke liye Calendar tool use karo (default 30 min).
- Kharche ke liye Expense tool use karo (amount, category, note).
- Calculation ke liye Calculator tool.
- Jawab chhota, Hinglish mein, emoji 1 max.
- Paise bhejne ya delete karne jaisa kaam kabhi mat karo — mana kar do.
```

💬 Build it with ChatGPT / Claude

📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Ek personal AI assistant Telegram par: "kal 4 baje Rahul ke saath meeting rakho", "₹450 lunch
expense likho", "18% GST of 12500?" — agent khud decide karta hai Calendar, Sheet ya Calculator kaunsa
tool use kare, aur pichhli baat yaad rakhta hai.

Flow: Telegram Trigger (Message) → AI Agent (Sochta + tool chunta) → Tools (Calendar · Sheet ·
Calculator) → Memory (Last 10 messages) → Telegram (Reply)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to
test it and the 3 most common errors beginners hit.
```

💬 Debug helper prompt

📋 Select karke copy karo

```
I'm building the n8n workflow "Telegram AI Assistant (Agent + Tools + Memory)" (nodes: Telegram Message,
AI Assistant, OpenAI Model, Chat Memory, Calculator, Log Expense, Book Meeting, Reply).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **QR scan karo** ya link kholo — workflow ki JSON file khulegi.
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ Workflow JSON
8 nodes · ready to import
adifyskills.com/workflows/day-15-telegram-ai-assistant.json

🌐 Lesson page
Copy buttons, quiz aur video links
adifyskills.com/day/15

Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Telegram bot token (BotFather se)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "OpenAI API key" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Google Sheets (Google OAuth)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Google Calendar (Google OAuth)" connect karo.
Time par apne aap nahi chala?	"Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo.
AI node par 401 / 429 error?	401 = API key galat. 429 = credit khatam ya bahut jaldi requests. OpenAI billing page check karo.
"Sheet not found" / column khaali?	Sheet ID aur tab ka naam (capital/small bhi) check karo. Pehli row mein headers hone chahiye.

⚠️ Yeh galtiyan sabse common hain

- ! Tool description vague — agent galat tool chunta hai
- ! Memory session key sabke liye same (sab users ki baatein mix)
- ! Agent ko delete/payment jaise risky tools bina approval

⭐ Pro tips — time bachane wali tricks

- ★ Tool ka naam aur description crystal clear: "book_meeting: books a calendar slot. Input: datetime, person"
- ★ Max iterations set karo — infinite loop se bacho
- ★ Risky actions par human-in-the-loop approval

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Telegram AI assistant jo meetings book kare, emails summarize kare aur expenses log kare.

📝 Homework

Apne niche ke liye ek agent banao jismein kam se kam 3 tools hon.

- ☐ Homework ho gaya

BEGINNER
Calculator tool wala agent

INTERMEDIATE
Sheet lookup tool: "mera order status?"

ADVANCED
Approval step before sending email

🧩 Aaj ka challenge — pehle khud try karo

🎯 Assistant ko ek naya tool do: "aaj ka total kharcha batao".

💡 **Hint 1**
Google Sheets Tool ko "read" operation ke saath bhi add kar sakte ho.

💡 **Hint 2**
Tool description saaf likho: "Expenses sheet padhta hai" — agent isi se decide karta hai.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Agent ko yaaddasht dene wala sub-node?

(A) Parser
(B) Memory
(C) Embeddings
(D) Splitter

Q2. Agent galat tool chun raha hai — fix?

(A) Temperature 2
(B) Kuch nahi
(C) Tool description improve
(D) Model delete

Q3. Multi-user bot mein session key?

(A) Fixed "1"
(B) Empty
(C) Date
(D) Har user ka unique ID

Q4. Workflow ko agent ka tool banane ka node?

(A) Call n8n Workflow Tool
(B) Merge
(C) Wait
(D) Sort

📖 Aaj ke naye shabd

Workflow
Steps ki ek chain jo apne aap chalti hai — jaise recipe ke steps.

Node
Workflow ka ek dabba/step. Har node ek kaam karta hai (email bhejna, data laana...).

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

API
Do apps ke baat karne ka tareeka. Ek app sawaal bhejta hai, doosra jawab (data) deta hai.

LLM
Large Language Model — ChatGPT/Claude jaisa AI jo text samajhta aur likhta hai.

Prompt
AI ko diya gaya instruction/sawaal.

Agent
AI jo khud decide karta hai kaunsa tool use karna hai — sirf baat nahi, kaam bhi karta hai.

Embedding
Text ka "matlab" numbers mein — taaki milte-julte text dhoondhe ja sakein.

✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 16

RAG: Chat With Your Own Data

"Company ke documents se jawab dene wala AI."

☐ **Day 15 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → B
Memory

Q2 → C
Tool description improve

Q3 → D
Har user ka unique ID

Q4 → A
Call n8n Workflow Tool

✅ Challenge ka solution

Naya Google Sheets Tool: operation Get Rows, sheet Expenses, description "Saare kharche padhta hai, total nikaalne ke liye". System prompt mein: "Total poocha jaaye to yeh tool use karo."

📚 Official docs

- AI Agent node ↗
- Memory ↗
- Chat Trigger ↗
- Call n8n Workflow Tool ↗
- AI agents explained ↗

▶ Videos

- n8n ai agent tutorial tools memory ↗
- n8n telegram ai agent ↗

💬 Doubt poochho

WhatsApp par seedha message

`wa.me/919000000000?text=Day%2015%20(AI%20Agents%3A%20Tools%2C%20Memory%20%26%20Reasoning)%20mein%20doubt%20hai%3A%20`

🌐 Day 15 lesson page

Quiz, copy buttons, videos

`adifyskills.com/day/15`